

Lecture 5. Fundamental limits in data compression

5.1 Introduction

So far, we have seen that Huffman codes are the optimal codes for any discrete source and that the average length L_{av} of such code is “close” to the entropy of the source. In other words, the optimal code for a source U taking values in $\{1, \dots, r\}$ with probability distribution $\{p_1, \dots, p_r\}$ satisfies

$$H(U) \leq L_{\text{av}} < H(U) + 1, \quad (5.1)$$

where $L_{\text{av}} = \sum_{j=1}^r \ell_j p_j$ is the average length of the code, and $H(U) = -\sum_{j=1}^r p_j \log_2 p_j$ is the entropy of the source. In the previous lecture, we found that the performance of the optimal code increased by compressing multiple symbols at once. That is, we can design our Huffman code for blocks of n symbols $X = (U_1, \dots, U_n)$ and get a code whose average length satisfies

$$H(X) \leq L_{\text{av}} < H(X) + 1. \quad (5.2)$$

Calculating the entropy $H(X)$ can be cumbersome, since the cardinality of the alphabet of X grows exponentially with n . That is, the number of possible source sequences of length n , given by n times the Cartesian product of the alphabet, is r^n . However, if the source is memoryless, $H(X)$ simplifies as lot.

5.2 Entropy of a discrete memoryless source

Consider a discrete source described by the sequence of random variables $X = (U_1, \dots, U_n)$ taking values in

$$\mathcal{X} = \mathcal{U}^n = \underbrace{\mathcal{U} \times \dots \times \mathcal{U}}_{n \text{ times}} \quad (5.3)$$

with joint distribution $P_X(x) = P_{U_1 \dots U_n}(u_1, \dots, u_n)$. The entropy of X is given by the expected value of the negative logarithm of the joint distribution, that is

$$H(X) = \mathbb{E} \left[\log_2 \frac{1}{P_X(X)} \right] = - \sum_{x \in \mathcal{X}} P_X(x) \log_2 P_X(x). \quad (5.4)$$

Since the source is memoryless, $X = (U_1, \dots, U_n)$ is a sequence of independent variables and therefore the distribution of X satisfies

$$P_X(x) = \prod_{i=1}^n P_{U_i}(u_i). \quad (5.5)$$

Using (5.5) in (5.4) we obtain that

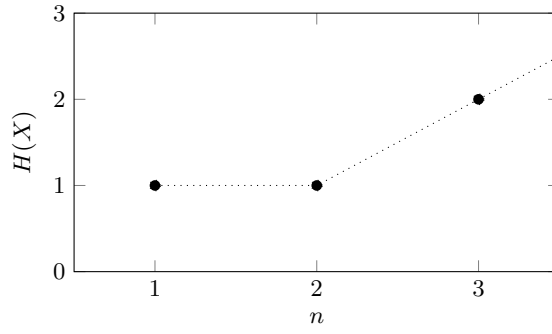
$$H(X) = -\mathbb{E} \left[\log_2 \prod_{i=1}^n P_{U_i}(U_i) \right] = -\mathbb{E} \left[\sum_{i=1}^n \log_2 P_{U_i}(U_i) \right] = \sum_{i=1}^n \mathbb{E} \left[\log_2 \frac{1}{P_{U_i}(U_i)} \right] = \sum_{i=1}^n H(U_i); \quad (5.6)$$

the entropy of a sequence of independent random variables becomes the sum of each individual entropy. Finally, since in a discrete memoryless source the symbols are identically distributed, all the entropies are the same and

$$H(X) = nH(U); \quad (5.7)$$

the entropy of a sequence of i.i.d. random variables is n times the entropy of a single symbol.

Problem 5.1. A discrete source $X = (U_1, \dots, U_n)$ outputs independent symbols U_i , all taking values over the alphabet $\{0, 1\}$. The entropy of X is given by the following plot as a function of n .



Which of the following statements is true?

- (a) The source is memoryless (U_i are independent and identically distributed for $i = 1, \dots, n$)
- (b) U_1 and U_3 have equiprobable symbols
- (c) $H(U_2) = 1$
- (d) The average length of any binary prefix-free code for X with $n = 3$ satisfies $L_{\text{av}} \geq 3$

Solution: Using (5.6) and from the plot, we infer that $H(U_1) = 1$, $H(U_1) + H(U_2) = 1$ and $H(U_1) + H(U_2) + H(U_3) = 2$, implying $H(U_2) = 0$ and $H(U_3) = 1$. Since $H(U_2) = 0$, the source cannot be memoryless (actually only if $H(U_1) = H(U_2) = H(U_3)$) invalidating (a) and (c). The entropy of X for $n = 3$ is $H(X) = 2$, therefore the average length of any binary prefix-free code for this source is lower bounded by 2, not 3. Finally, (b) is true because both U_1 and U_3 take values in $\{0, 1\}$ and, for equiprobable bits, both $H(U_1)$ and $H(U_3)$ are indeed equal to 1.

5.3 Compressing an information source

We are now equipped to show the limiting performance of compressing an information source X with the optimal (Huffman) binary prefix-free code as the size of the encoded source blocks n tends to infinity. To do so, we define the average length per symbol as

$$L_n = \frac{L_{\text{av}}}{n}, \quad (5.8)$$

where L_{av} is the average length of the optimal (Huffman) code for a discrete memoryless source $X = (U_1, \dots, U_n)$. Similarly, we define the efficiency of a prefix-free code used to represent a source X of length n as

$$e_n = \frac{H(X)}{L_{\text{av}}}. \quad (5.9)$$

If the source is memoryless, then $H(X) = nH(U)$ and the efficiency can be written using the average length per symbol L_n in (5.8) as

$$e_n = \frac{nH(U)}{L_{\text{av}}} = \frac{nH(U)}{nL_n} = \frac{H(U)}{L_n}. \quad (5.10)$$

Example 5.2. Consider the discrete memoryless source $X = (U_1, \dots, U_n)$ where each U_i takes values in $\{a, b\}$ with probability $\{\frac{2}{3}, \frac{1}{3}\}$. For $n = 1$, we have $H(X) = H(U) = \log_2(3) - \frac{2}{3}$ bits, while the optimal prefix-free code has length distribution $\{1, 1\}$ leading to $L_{\text{av}} = 1$ bit. Using equations (5.8), and (5.10) we obtain $L_1 = 1$ and an efficiency of

$$e_1 = \frac{H(X)}{L_{\text{av}}} = \frac{H(U)}{L_1} = H(U) = \log_2(3) - \frac{2}{3} \approx 0.918296. \quad (5.11)$$

For $n = 2$, we have that X takes values in $\{aa, ab, ba, bb\}$ with probability $\{\frac{4}{9}, \frac{2}{9}, \frac{2}{9}, \frac{1}{9}\}$, hence an entropy of $H(X) = 2H(U) = 2\log_2(3) - \frac{4}{3}$. The optimal prefix-free code for X is, for example, the Huffman code $\{0, 10, 110, 111\}$ of length distribution $\{1, 2, 3, 3\}$, giving an average length of $L_{\text{av}} = \frac{17}{9}$. Since we are encoding

blocks of $n = 2$ symbols, the average length per symbol in this case is $L_2 = \frac{L_{\text{av}}}{2} = \frac{17}{18}$ bits per symbol. With these numbers, we obtain an efficiency of

$$e_2 = \frac{H(X)}{L_{\text{av}}} = \frac{2 \log_2(3) - \frac{4}{3}}{17/9} \approx 0.972313, \quad (5.12)$$

slightly higher than the one in (5.11).

Problem 5.3. Calculate the efficiency of the optimal prefix-free code for X in Example 5.2 for $n = 3$.

Solution: To determine the efficiency e_3 we need the entropy of X from the one hand, and the average length of the Huffman code for X from the other. Since the source is memoryless, we do not need to calculate the whole alphabet and distribution of X to calculate the entropy and simply use that $H(X) = 3H(U) = 3 \log_2(3) - 2$. To construct the Huffman code, we need to determine the alphabet and distribution of X . After doing so, building the Huffman tree and finding the length distribution (left as exercise!), we obtain an average length of $L_{\text{av}} = \frac{76}{27}$. Hence,

$$e_3 = \frac{H(X)}{L_{\text{av}}} \approx 0.97871, \quad (5.13)$$

another increase of efficiency with respect to (5.12).

Problem 5.4. We design a sub-optimal prefix-free code for the discrete memoryless source X in Example 5.2. For any n , the lengths of such code satisfy $\ell(x) = \lceil -\log_2 P_X(x) \rceil$. Calculate the efficiency of this code for $n = 3$ and compare it with that of the optimal code.

Solution: For $n = 3$, X takes values in $\{aaa, aab, aba, abb, baa, bab, bba, bbb\}$ with probability distribution $\{\frac{8}{27}, \frac{4}{27}, \frac{4}{27}, \frac{2}{27}, \frac{4}{27}, \frac{2}{27}, \frac{2}{27}, \frac{1}{27}\}$. The efficiency of the optimal code was already calculated in (5.13). The sub-optimal code will encode $x = aaa$ with $\ell(x) = \ell(aaa) = \lceil -\log_2 P_X(aaa) \rceil = \lceil -\log_2 \frac{8}{27} \rceil \approx \lceil 1.75489 \rceil = 2$ bits. With the same procedure we obtain a length distribution $\{2, 3, 3, 4, 3, 4, 4, 5\}$, giving an average length of $L_{\text{av}} = 3$ and an efficiency of $e_3 \approx 0.918296$, smaller than the optimal one.

We have seen from the previous exercises that increasing n may increase the efficiency, meaning that L_{av} approaches $H(X)$ or that L_n approaches $H(U)$. To see this, we combine the results obtained in this section with those in Section 5.1. Using (5.8) and (5.7) back in (5.2), we have that the average length per symbol of the optimal prefix-free code for a discrete memoryless source X satisfies (after dividing by n all the terms):

$$H(U) \leq L_n < H(U) + \frac{1}{n}. \quad (5.14)$$

Since L_n is lower and upper bounded as in (5.14), we conclude that

$$\lim_{n \rightarrow \infty} L_n = H(U); \quad (5.15)$$

the compression of blocks is very efficient as the corresponding average length per symbol L_n converges to the entropy of the source. The same result can be expressed in terms of the efficiency. Since the average length of any prefix-free code for X is lower bounded as $L_{\text{av}} \geq H(X)$, from the definition of the efficiency in (5.10) we find that the efficiency is upper bounded as $e_n \leq \frac{H(X)}{H(X)} = 1$. Similarly, using that the average length of the optimal code satisfies $L_{\text{av}} < H(X) + 1$, we have that $e_n > \frac{H(X)}{H(X)+1}$. Combining everything, we have

$$\frac{H(X)}{H(X)+1} < e_n \leq 1. \quad (5.16)$$

If the source is memoryless, then (5.16) can be written as

$$\frac{H(U)}{H(U) + \frac{1}{n}} < e_n \leq 1, \quad (5.17)$$

implying that

$$\lim_{n \rightarrow \infty} e_n = 1. \quad (5.18)$$

Example 5.5. Consider a discrete memoryless source $X = \{U_1, \dots, U_n\}$ where each U_i takes values in $\{a, b, c\}$ with probability $\{\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\}$. The entropy of U is $H(U) = \frac{3}{2}$. For example, for $n = 10$, this implies that the average length of the optimal code for X satisfies $15 \leq L_{\text{av}} < 16$, or that the average length per symbol satisfies $\frac{3}{2} \leq L_{10} < \frac{8}{5}$, or that the efficiency satisfies $\frac{15}{16} < e_{10} \leq 1$.

We end the last lecture about source coding with a problem relating probability and the limiting performance of compressing an information source.

Problem 5.6. A discrete memoryless source $X = (U_1, \dots, U_n)$ has probability distribution $P_X(x)$. We define the random variable $Y = f(X)$, where $f(x) = -\log_2 P_X(x)$. Calculate $\lim_{n \rightarrow \infty} \mathbb{P}[Y < n^2 H(U)]$.

Solution: Using Markov's inequality, since Y is a non-negative random variable, we have that

$$\mathbb{P}[Y < n^2 H(U)] = 1 - \mathbb{P}[Y \geq n^2 H(U)] \geq 1 - \frac{\mathbb{E}[Y]}{n^2 H(U)}. \quad (5.19)$$

The expected value of Y can be calculated in terms of X , giving $\mathbb{E}[Y] = \mathbb{E}[f(X)] = \mathbb{E}[-\log_2 P_X(X)]$, which is the definition of the entropy of X according to (5.4). Hence, $\mathbb{E}[Y] = H(X)$. Since the source is memoryless, from (5.7), we also have that $H(X) = nH(U)$. Combining all the results, we obtain

$$1 \geq \mathbb{P}[Y < n^2 H(U)] \geq 1 - \frac{1}{n}, \quad (5.20)$$

where the first bound is a trivial bound. Taking the limit as $n \rightarrow \infty$ to all terms above, we conclude that $\lim_{n \rightarrow \infty} \mathbb{P}[Y < n^2 H(U)] = 1$.